# Natural Language Processing

Coursework Report

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GitHub Repository Link

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# 1 Exercise 1

## 1.1 Primary Contributions

One of the main primary contributions of the paper is proposing a new dataset, containing 10,000 paragraphs from news articles across 20 countries that is tailored to identifying PCL, specifically targetted towards vulnerable communities. As compared to hate speech or offensive language which has been substantially researched on NLP, it proposed a dataset that is still in an emerging area while having a clear goal to reduce exclusion and inequalities between these communities.

Another primary contribution is the two-step human annotation in annotating the two-level categories of PCL from paragraphs, by proposing the detection of PCL into a binary classification and a multi-label classification problem. The annotation provides high-quality training data to the NLP models.

A final primary contribution is providing the performance metrics of SVM, bidirectional LSTM, and fine-tuned language models including different variants of BERT. The precision, recall and F1 values are provided in the paper, as well as potential reasons why some models perform more poorly in some categories compared to others. This provides a reference baseline and allows substantial future work to be done to justify these patterns.

## 1.2 Technical Strengths

One technical strength is dividing the detection of PCL into two tasks: a binary classification task (Task 1) and a multi-label classification task (Task 2), with rigorously designed labels. In the first task, the labels are made fair and unbiased by computing the IAA between two annotators, with a clear definition to handle borderline cases between them by setting a threshold for the sum of the labels. In the second task, a detailed 2 level PCL categories definition is used, with example given for each definition. Each paragraph is assigned a subset of the PCL labels instead of a single label, allowing multi-label classification. The carefully designed way to label data allowed the training data to be as trustworthy and as unbiased as possible.

Another technical strength is the comprehensive model metrics computed for different approaches compared to the random baseline, obtained from 10-fold cross validation. The paper displayed the metrics per method per category, allowing conclusions to be drawn and are easily understandable by the readers, such as 'the poorer, the merrier' having the lowest performance across all categories. It also provided examples of wrong prediction, and identified structure of languages that may lead to this.

## 1.3 Key Weaknesses and Improvements

One key weakness is its lack of explanation before drawing conclusions for poor performance. It mentioned that BERT-large has surprisingly poor performance compared to other BERT models, as a result of overfitting due to the small number of training examples but a deeper architecture. However, it does not provide evidence to justify that overfitting is happening, such as showing the train / validation loss.

The second weakness is the imbalanced number of training examples. It attributed the poor performance of 'the poorer, the merrier' to the lower number of training examples, but this can be mitigated by having a better balance between classes during data

gathering. It would then be hard to verify if poor performance is due to the nature of the category, or the imbalanced number of training examples.

# 2 Exercise 2

## 2.1 Univariate Analysis

### 2.1.1 Impact Statement

Univariate analysis discovers noisy or missing data in order to determine if the training data needs to be cleaned. Also, this part discovers the distribution of keywords, sentence lengths and N-grams, to identify class imbalances and outliers. This would help estimate the number of parameters, and identify if the size of certain classes of specific keywords need to be inflated through oversampling.

### 2.1.2 Duplicates and Noisy Data

One of the training set rows is missing the value of text, since there is only a single row, that row was removed during training.

Some text sentences contain special characters, where the characters were removed. Table 1 shows the special characters and the number of rows containing them.

| Special Character | Count |
|---|---|
| `<h>` | 401 |
| `&` | 61 |
| `>` | 17 |
| `<` | 0 |
| `\n` | 0 |
| `\t` | 0 |

Table 1: Frequencies for special characters

### 2.1.3 Keywords Distribution

Table 2 shows the distribution of all unique keywords from the keywords column. All keywords have a similar count except the count of 'poor-families' keyword is around 100 less.

| Keyword | Count |
|---|---|
| migrant | 1088 |
| in-need | 1082 |
| vulnerable | 1080 |
| homeless | 1077 |
| women | 1070 |
| refugee | 1068 |
| immigrant | 1061 |
| disabled | 1028 |
| hopeless | 1005 |
| poor-families | 909 |

Table 2: Keywords distribution

#### 2.1.4 Sentence Length Distribution

Figure 1 shows the distribution of the number of words in the text. Table 3 also shows the metrics of the number of words in the text. There are outliers in the number of words, with 95% of sentence length falling in between the range of $[11, 119]$.

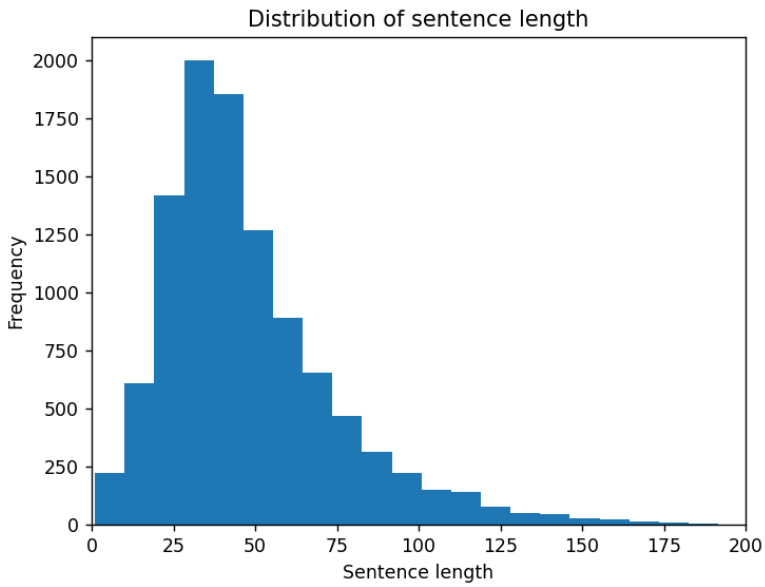


Figure 1: Number of words distribution

| Metric | Value |
|---|---|
| min | 1 |
| max | 909 |
| mean | 48.43 |
| median | 42 |

Table 3: Sentence length metrics

#### 2.1.5 N-grams Analysis

Table 4 and 5 show the top 10 most frequent bigrams and trigrams in all text respectively, excluding n-grams that contains stop words.

| **Bigram** | **Count** |
|---|---|
| poor families | 909 |
| illegal immigrants | 307 |
| per cent | 165 |
| united states | 161 |
| homeless people | 136 |
| last year | 128 |
| disabled people | 116 |
| sri lanka | 108 |
| new york | 104 |
| hong kong | 102 |

Table 4: Top 10 most frequent bigrams

| **Trigram** | **Count** |
|---|---|
| president donald trump | 55 |
| un refugee agency | 31 |
| many poor families | 28 |
| us president donald | 19 |
| dar es salaam | 17 |
| world war ii | 17 |
| president barack obama | 17 |
| help poor families | 16 |
| urban poor families | 14 |
| working poor families | 14 |

Table 5: Top 10 most frequent trigrams

### 2.2 Bivariate Analysis

#### 2.2.1 Impact Statement

Bivariate analysis discovers the relation between two different columns. This is important as analysing label distribution by keywords allow us to determine if the result of a certain class may be due to its class imbalance, and logistic regression allow us to determine the most important n-grams that impact the label, so we would know if the n-grams align with the task and determine what should be focused on.

#### 2.2.2 Label Distribution By Keywords

Figure 2 shows the distribution of labels (0 to 4) for each keyword class. It is observed that labels 0 and 1 corresponding to not PCL occupies the majority of data, with some particular keywords such as 'immigrant' having 97.2% of text classified as not PCL.

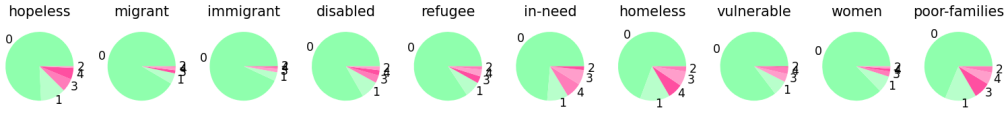


Figure 2: Label distribution by keywords

#### 2.2.3 Logistic Regression of Bigrams and Trigrams

A logistic regression was performed to investigate which bigram or trigram leads to the highest expected change in the likelihood of positive PCL, according to the coefficients of the regression model. Table 6 shows the top 10 bigrams and trigrams that most likely increases the likelihood of PCL classification to be True and False respectively, in descending order.

It was discovered that the bigrams that lead to True prediction typically contains an adjective describing a person (homeless, young, rough, disabled).

| True Prediction | False Prediction |
|---|---|
| homeless person | illegal immigrants |
| pope francis | many poor |
| disabled man | anti immigrant |
| come true | family members |
| young man | hopeless romantic |
| rohingya refugees | long term |
| sleeping rough | social protection |
| let alone | non profit |
| less fortunate | new zealand |
| two men | people homeless |

Table 6: Bigrams and trigrams with the most impact on the likelihood of classification

# 3 Exercise 3

## 3.1 Proposed Approach

The proposed approach consists of 4 modifications: Text cleaning, oversampling, adding contextual information and using model ensembles.

#### 3.1.1 Text Cleaning

The first step is text cleaning that involves removing the special characters in table 1, and replacing all numbers in the text with 0. As these information add little meaning to the text data, they were removed so the model would focus on interpreting text semantics.

There is also one row without text value, which was removed during training and validation.

### 3.1.2 Oversample the Minority Class

For each keyword, sampling with replacement was used to oversample the minority (positive) class in the training dataset for each corresponding keyword, until the positive class occupies a certain percentage of the data. The value chosen here was 25.

Table 3.1.2 shows the number of positive and negative training examples before and after oversampling.

| | **True labels** | **False labels** |
|---|---|---|
| Before oversampling | 794 | 7581 |
| After oversampling | 2527 | 7581 |

### 3.1.3 Add Contextual Information

The keyword and the country code associated with the text was prepended to the text before feeding it into the model, in the following format.

```
in need<SEP>nz<SEP>More than 750 trolleys of food donated to those in need
```

The hyphens in "in-need" and "poor-families" were replaced by spaces to match the format found in the text.

### 3.1.4 Use of Model Ensembles

Using the techniques described above, a separate BERT model was trained in the same way on the same text data. An ensemble technique, which combines the output of both the BERT and RoBERTa model was used, by assigning different weights to each of them described in the equation below.

$$predictions = argmax(x * RoBERTa + (1 - x) * BERT) \tag{1}$$

$RoBERTa$ and $BERT$ refers to the predicted probabilities of class 0 and 1, for each of the models, respectively. The weight, $x$, was found using the grid search technique to be 0.8.

## 3.2 Rationale and Expected Outcome

### 3.2.1 Oversampling

As shown in the analysis, there were huge class imbalances for each keyword, where some keywords have a lower proportion of positive labels than others. This may encourage the model to predict every text as negative during training, as it sees much more negative examples than positive ones. By artificially enlarging the size of the positive rows, the model is expected to make less conservative decisions, leading to a higher recall value.

### 3.2.2 Adding Contextual Information

The paper mentioned that categories with the lowest performance typically require some forms of world knowledge[1]. However, from the given dataset, the only form of external social context information is the keyword and country code.

In an experiment to classify fake news related to COVID-19 on Twitter, it was shown that including user and Tweet socio-contextual Information such as user description, verified status, and the number of likes, helped the model to achieve better results, where the user information was particularly valuable[2]. Based on these findings, the keyword and country code was added to the language models in the same way proposed in this paper, and since the target is to detect the use of PCL specifically towards vulnerable communities, the inclusion of keywords may help the model to better understand what it should focus on. The inclusion of country codes may also help the model to understand the different tones and contexts used in each country.

#### 3.2.3 Model Ensembles

In an experiment of classifying idioms in sentences, the use of weighted average of a BERT model and RoBERTa model performs better than each single individual model, in order to reduce noise, bias and variance in model predictions, which are major causes of errors[3]. A similar technique was implemented here to combine the strengths of both models, and their combined F1 score is expected to be higher than each of their individual ones.

# 4 Exercise 5

## 4.1 Error Analysis

#### 4.1.1 Error Samples of the Final Model

Table 7 shows some examples of false positives (FP) and false negatives (FN) according to the predictions of the final model.

| **Sentences** | **FP / FN** |
|---|---|
| Bombarded by schizophrenia , addiction and homelessness , you might say that Eoghan O'Driscoll has been to hell and back . But he is finding a new balance through painting . Interview : Michael Lanigan | FN |
| A crew of disabled athletes will be tackling this weekend 's Chattanooga Waterfront Triathlon to show others with disabilities they , too , can participate in a healthy , active lifestyle . | FP |
| Her fate was to end up drowning herself in a bog because she would n't conform as women were expected to . | FP |
| Many refugees do n't want to be resettled anywhere , let alone in the US . | FN |

Table 7: Examples of misclassified sentences

The misclassified samples show that even with the improved model, it still struggles to detect certain phrases, and is still weak at understanding the context of the text. In the first sentence, although having a label of 4 with the high agreement from the annotators, the model predicted a false negative. This could be attributed to the use of metaphors in "been to hell and back", which the model may struggle to understand. In the second sentence, although words such as "too" are used to describe the disabled, the model failed to identify that this is not used in a condescending context, and instead is

in a more descriptive tone. This shows that although suitable improvements have been made to the model, it still shows the same weaknesses described in the original paper.

#### 4.1.2 Comparison with the Baseline

Compared to the baseline model, the validation F1 score of the proposed approach improved from 0.577 to 0.630. Tables 8, 9 and 10 show the F1, precision and recall validation values for both the baseline model and final model, classified into each keyword. It appears that certain keywords are more difficult to classify and hence the lower F1 scores, namely women and migrant, which can be explained by them having some of the least number of positive samples from the original dataset. It is also observed that the precision score dropped for some keywords, but the recall increased for most keywords, indicating the model is making less conservative predictions, and is less affected by the class imbalances.

| **Keyword** | **Baseline** | **Final** |
|---|---|---|
| Overall | 0.577 | 0.630 |
| women | 0.286 | 0.414 |
| in-need | 0.761 | 0.780 |
| immigrant | 0.600 | 0.769 |
| hopeless | 0.509 | 0.613 |
| homeless | 0.613 | 0.698 |
| vulnerable | 0.537 | 0.625 |
| migrant | 0.500 | 0.400 |
| disabled | 0.615 | 0.485 |
| poor-families | 0.560 | 0.585 |
| refugee | 0.462 | 0.625 |

Table 8: F1 Scores compared to baseline

| **Keyword** | **Baseline** | **Final** |
|---|---|---|
| Overall | 0.582 | 0.561 |
| women | 0.429 | 0.400 |
| in-need | 0.711 | 0.653 |
| immigrant | 1.000 | 0.833 |
| hopeless | 0.483 | 0.528 |
| homeless | 0.576 | 0.647 |
| vulnerable | 0.524 | 0.536 |
| migrant | 0.667 | 0.400 |
| disabled | 0.667 | 0.421 |
| poor-families | 0.568 | 0.545 |
| refugee | 0.462 | 0.526 |

Table 9: Precision compared to baseline

| Keyword | Baseline | Final |
|---|---|---|
| Overall | 0.573 | 0.719 |
| women | 0.214 | 0.429 |
| in-need | 0.818 | 0.970 |
| immigrant | 0.429 | 0.714 |
| hopeless | 0.538 | 0.731 |
| homeless | 0.655 | 0.759 |
| vulnerable | 0.550 | 0.750 |
| migrant | 0.400 | 0.400 |
| disabled | 0.571 | 0.571 |
| poor-families | 0.553 | 0.632 |
| refugee | 0.462 | 0.769 |

Table 10: Recall compared to baseline

## 4.2 Ablation Studies

### 4.2.1 Only Oversample

Table 11 shows the comparison of F1, precision and recall scores of performing only oversampling on the baseline model with the baseline model itself.

| | Baseline | Only Oversample |
|---|---|---|
| F1 score | 0.577 | 0.587 |
| Precision | 0.582 | 0.559 |
| Recall | 0.573 | 0.618 |

Table 11: Metrics for baseline vs only oversampling

It was observed that after oversampling the positive minority class, the precision dropped but the recall increased, leading to an overall improvement of the F1 score. This aligns with the expected results, as the model after oversampling is less likely to bias towards negative predictions because most of the training data are negative.

### 4.2.2 Oversample and Adding Context

Table 12 shows the F1 score for each keyword, compared between the only oversample model and oversample with context model.

| Keyword | Oversample | Oversample with Context |
|---|---|---|
| Overall | 0.587 | 0.619 |
| women | 0.538 | 0.400 |
| in-need | 0.730 | 0.762 |
| immigrant | 0.364 | 0.769 |
| hopeless | 0.557 | 0.594 |
| homeless | 0.613 | 0.677 |
| vulnerable | 0.591 | 0.625 |
| migrant | 0.600 | 0.500 |
| disabled | 0.414 | 0.471 |
| poor-families | 0.528 | 0.588 |
| refugee | 0.667 | 0.606 |

Table 12: F1 score for oversample vs oversample with context

With the inclusion of contextual information including the keyword and country code, the model is able to perform much better on certain keywords such as immigrant and vulnerable. However, the performance also dropped on some keywords such as women and migrant. This shows that while adding these information may let it understand certain keywords better, the performance improvements are not obvious.

#### 4.2.3 Coreference Resolution

An attempt has been made using coreference resolution, by replacing pronouns or references with the actual entity name, using the `fastcoref` module. Combined with the inclusion of keywords, it is expected to allow the model to better recognise the main subject better in order to detect PCL specifically towards them. An example is shown below.

```
Before:
Dr. Lester Keith , doctor and professor of business administration , and
others are checking with local transportation groups to see if they can bring
those in need of a meal to the college for the 4 p.m. dinner . We will also
be contacting local soup kitchens as a pickup location and will work with
them to transport any leftovers to them so there is no wasted food , Dr. Keith
said .
After:
Dr. Lester Keith , doctor and professor of business administration , and
others are checking with local transportation groups to see if local transportation
groups can bring those in need of a meal to the college for the 4 p.m. dinner
. Dr. Lester Keith , doctor and professor of business administration , and
others will also be contacting local soup kitchens as a pickup location and
will work with local soup kitchens to transport any leftovers to local soup
kitchens so there is no wasted food , Dr. Lester Keith , doctor and professor
of business administration said .
```

Table 13 shows the performance of the model after oversampling and adding context, then with and without coreference resolution (CR), respectively.

| | Without CR | With CR |
|---|---|---|
| F1 score | 0.619 | 0.565 |
| Precision | 0.541 | 0.536 |
| Recall | 0.724 | 0.598 |

Table 13: Metrics for CR vs no CR

Surprisingly, the model performed noticeably worse when text were modified this way, and it predicted sentences wrong that were previously correct, even for those that remained unchanged after the modification. This can be explained by the RoBERTa model not being trained on natural language of this format, causing it to understand the text context poorly.

#### 4.2.4 Ensemble Model

Table 14 shows the number of false and true validation samples where the ensemble model predicts correctly, but the improved RoBERTa model or the improved BERT model fails to predict.

| | RoBERTa Fails | BERT Fails |
|---|---|---|
| False samples | 12 | 29 |
| True samples | 1 | 44 |

Table 14: Number of samples predicted correctly by ensemble, but incorrectly by improved RoBERTa or BERT

This shows that RoBERTa missed more false samples, while BERT missed more true samples, indicating that the BERT model may be more conservative in its predictions. This also shows that the ensemble approach was effective, by combining the strengths of both models.

## References


[1] Carla Perez-Almendros et. al. Don't patronize me! an annotated dataset with patronizing and condescending language towards vulnerable communities.

[2] Rishabh Upadhyay et. al. Leveraging socio-contextual information in bert for fake-health news detection in social media.

[3] J Briskilal and C.N. Subalalitha. An ensemble model for classifying idioms and literal texts using bert and roberta. *Information Processing Management*, 59(1):102756, 2022.